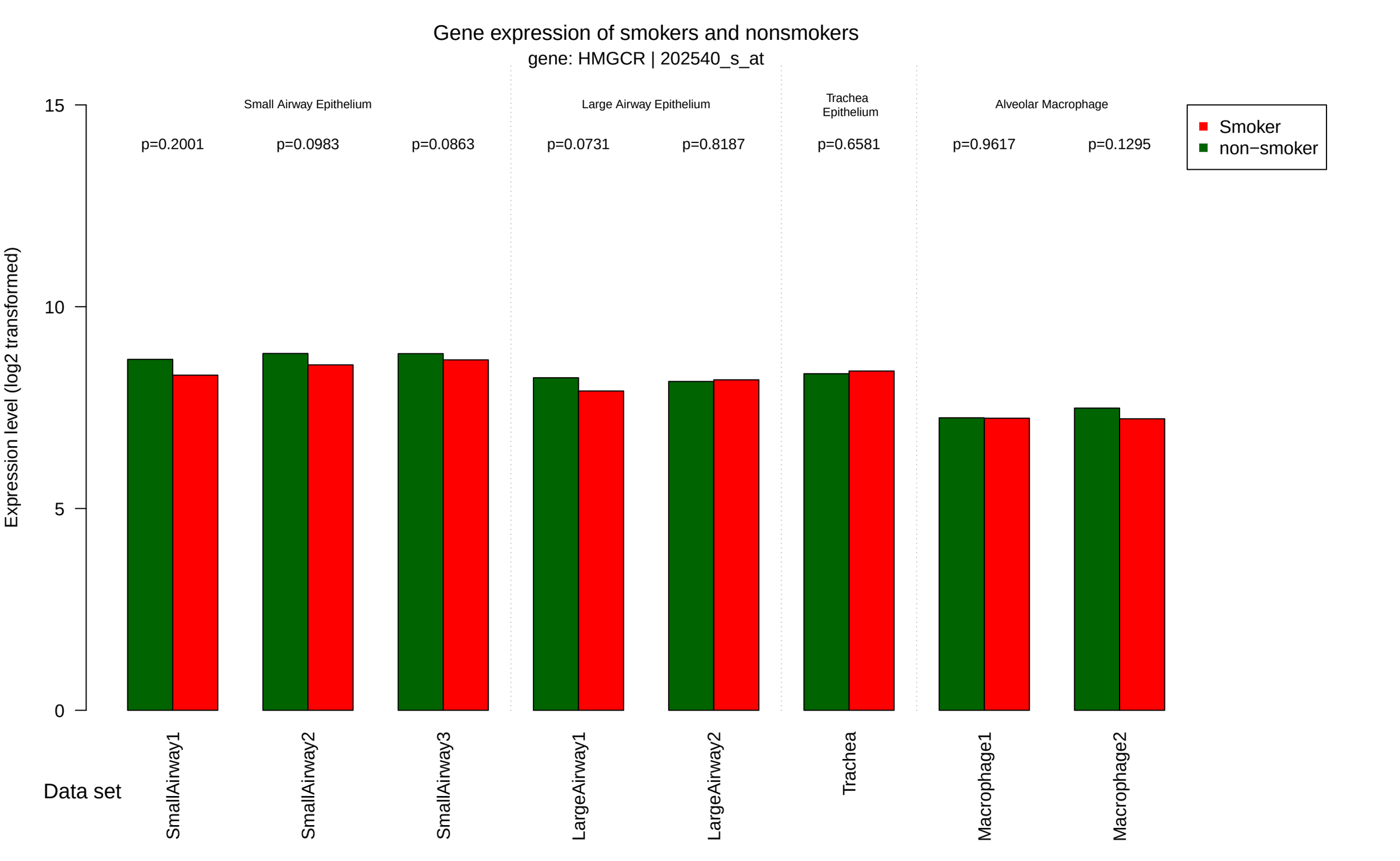
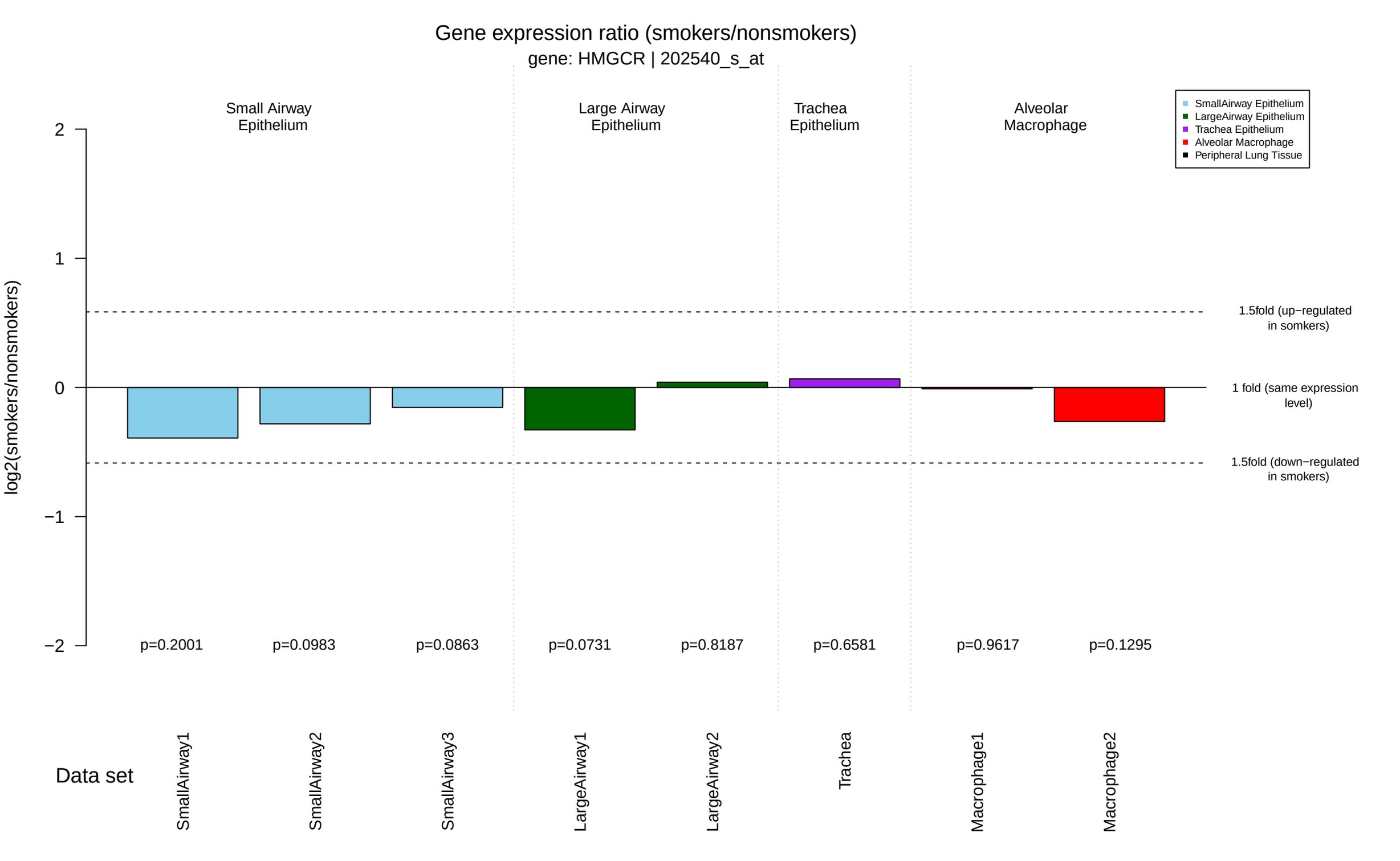
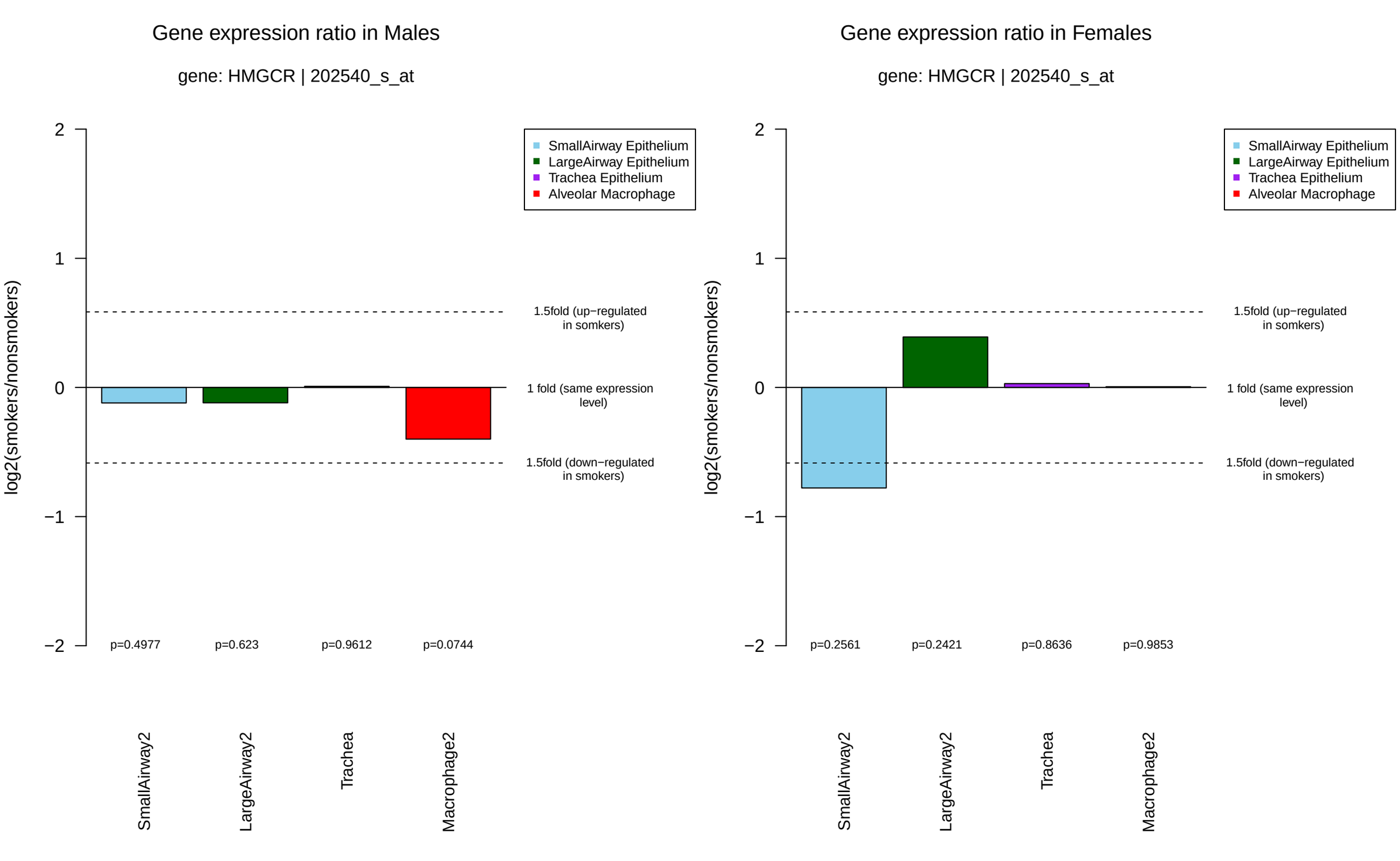
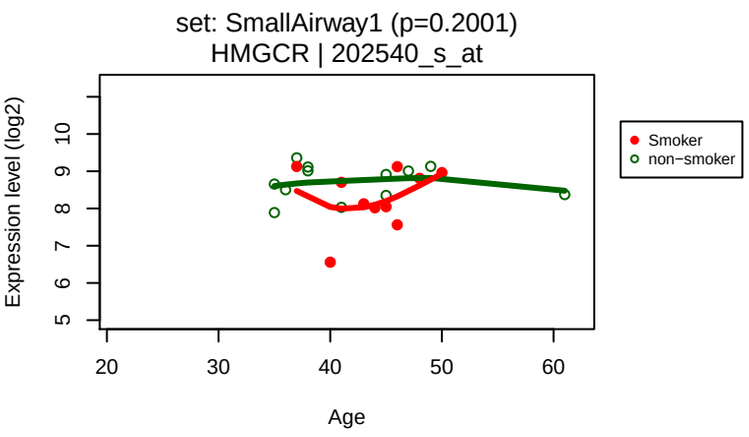
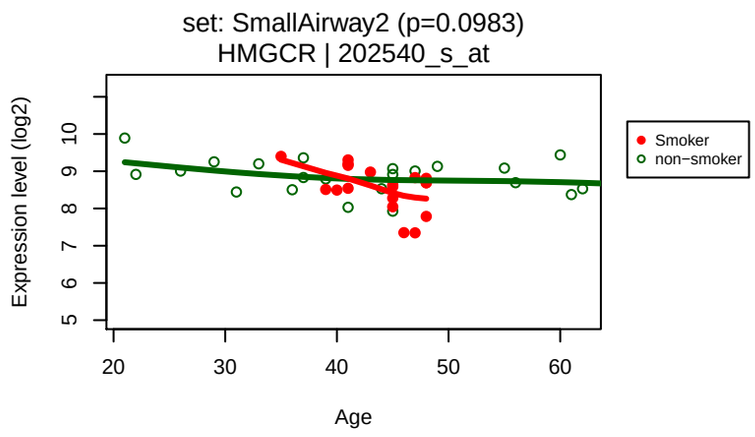
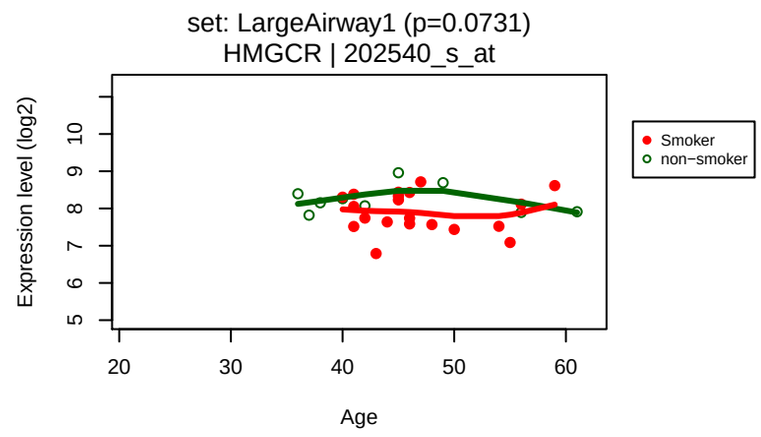

Gene expression of smokers and nonsmokers
gene: HMGCR | 202540_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2001
p=0.0983
p=0.0863
p=0.0731
p=0.8187
p=0.6581
p=0.9617
p=0.1295
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

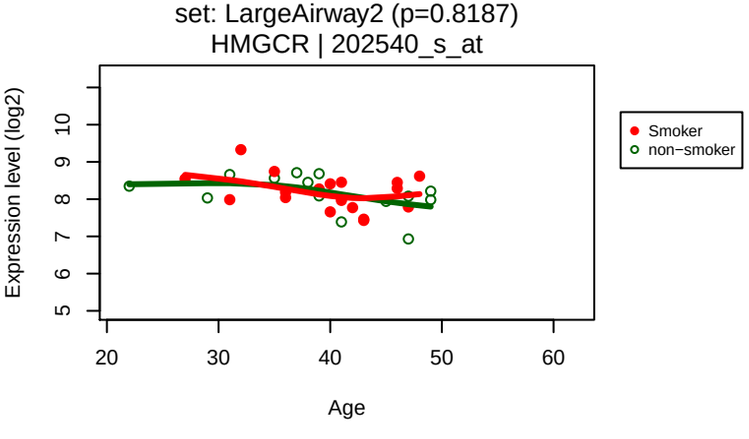

Gene expression ratio (smokers/nonsmokers)
gene: HMGCR | 202540_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2001
p=0.0983
p=0.0863
p=0.0731
p=0.8187
p=0.6581
p=0.9617
p=0.1295
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

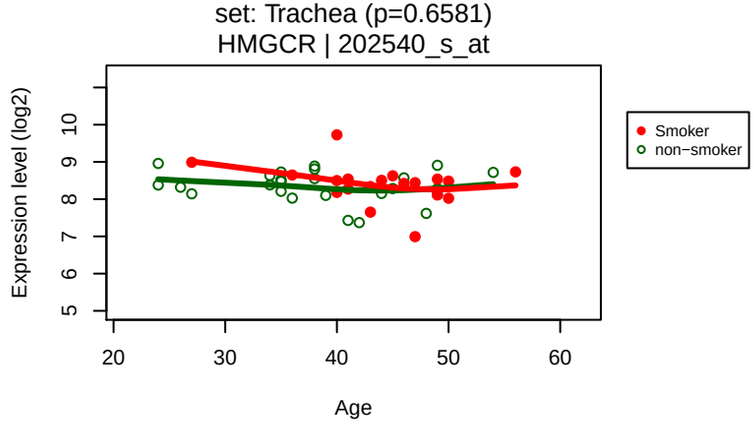

Gene expression ratio in Males
gene: HMGCR | 202540_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4977
p=0.623
p=0.9612
p=0.0744
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HMGCR | 202540_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2561
p=0.2421
p=0.8636
p=0.9853
SmallAirway2
LargeAirway2
Trachea
Macrophage2

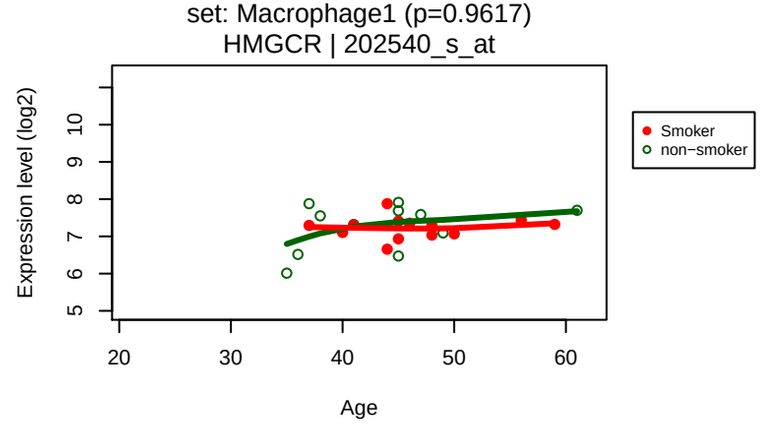

set: SmallAirway1 (p=0.2001)
HMGCR | 202540_s_at
Expression level (log2)
Age
Smoker
non−smoker

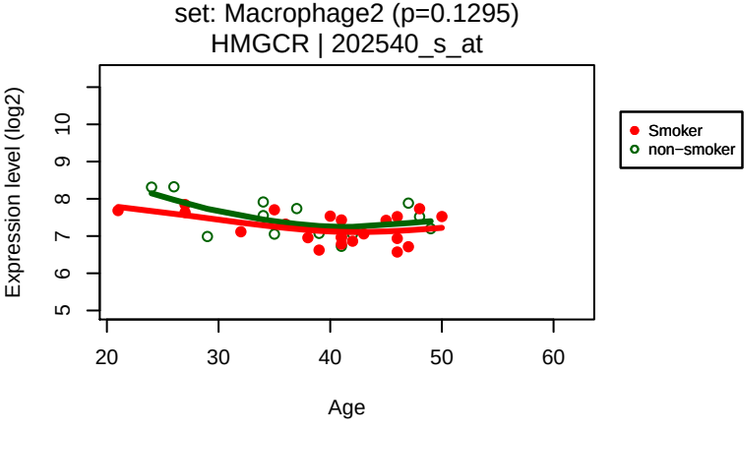

set: SmallAirway2 (p=0.0983)
HMGCR | 202540_s_at
Expression level (log2)
Age
Smoker
non−smoker

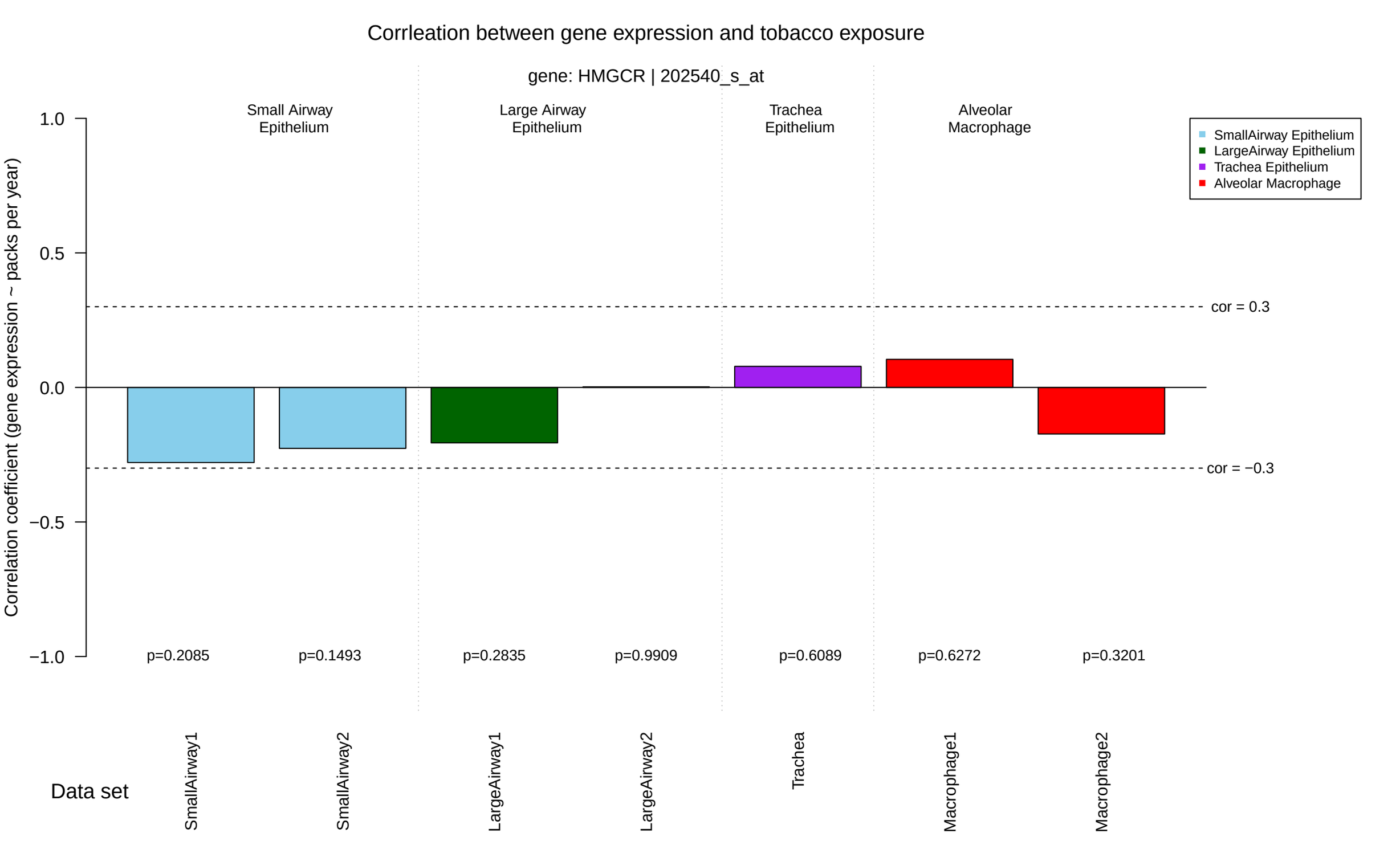

set: LargeAirway1 (p=0.0731)
HMGCR | 202540_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8187)
HMGCR | 202540_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6581)
HMGCR | 202540_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9617)
HMGCR | 202540_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1295)
HMGCR | 202540_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HMGCR | 202540_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2085
p=0.1493
p=0.2835
p=0.9909
p=0.6089
p=0.6272
p=0.3201
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2